

## Multiple Choice

1. **Answer:** C

*Options:* A) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent; B) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent; C) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent; D) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent; E) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent

*Note:* Correct option: C - Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent

2. **Answer:** A

*Options:* A) -28; B) 28; C) -7; D) 0; E) 7

*Note:* Correct option: A - -28

3. **Answer:** A

*Options:* A) majority carriers; B) dominant carriers; C) neutral carriers; D) non-conductive carriers; E) charge carriers

*Note:* Correct option: A - majority carriers

4. **Answer:** A

*Options:* A)  $3/2$ ; B)  $1/4$ ; C) 0; D)  $1/5$ ; E)  $1/3$

*Note:* Correct option: A -  $3/2$

5. **Answer:** A

*Options:* A) 660 watts; B) 880 watts; C) 200 watts; D) 120 watts; E) 110 watts

*Note:* Correct option: A - 660 watts

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: ' $F(t) = - (1/2) t^2 e^{-t} + 3te^{-t} - 4e^{-t} + te^{-2t} + 4e^{-2t}$ '.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'Negative'.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: ' $\{Q^2 / (16\pi\epsilon_0 R)\}$ '.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '0.625'.

## Long Form Response

11. **Answer:** 0.3 mm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:**  $3.5 \times 10^8$  meters/sec. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*